

MACHINE LEARNING

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PROJECT : MACHINE LEARNING ASSIGNMENT I

**TOPICS : Car Price Prediction - Multiple Linear
Regression AND Bank Customer Churn Prediction -
Logistic Regression**

TASK 2

Car Price Prediction - Multiple Linear Regression

I Domain Explanation

The automobile industry depends on several factors when determining the price of a vehicle. Features such as engine size, horsepower, dimensions, fuel efficiency, body type, and manufacturer can all affect the final price.

In this study, Multiple Linear Regression is used to predict car prices based on these features. EDA is first carried out to understand the data and identify useful relationships before developing and evaluating the model.

II Problem Statement

The aim of this study is to build a Multiple Linear Regression model to predict automobile prices using the available vehicle features.

The study focuses on identifying the factors that influence price, selecting relevant features, evaluating the model using MAE, MSE, RMSE, and R^2 , and checking important regression assumptions such as multicollinearity and heteroscedasticity. The baseline and reduced models are also compared to determine their performance.

Target Variable: price

III Dataset Explanation

The Car Price Prediction dataset contains information about 205 automobiles and 26 variables describing their technical and physical characteristics. The variables include both numerical and categorical features, such as engine size, horsepower, curb weight, fuel type, body type, drive wheel, and manufacturer.

The variable price is used as the target variable, while the remaining relevant features are considered as predictors. The dataset also contains 147 unique car names, so CarName was converted into a manufacturer feature to make it more useful for analysis.

Dataset Summary

Category	Details
Number of records	205
Number of variables	26
Target variable	price
Variable types	Numerical and categorical
Unique car names	147

The dataset contains a combination of numerical and categorical automobile characteristics, with price used as the target variable for prediction.

IV Exploratory Data Analysis (EDA)

4.1 Descriptive Statistics

Statistical summary of numerical variables

df.describe().T

	count	mean	std	min	25%	50%	75%	max
car_ID	205.000	103.000	59.323	1.000	52.000	103.000	154.000	205.000
symboling	205.000	0.834	1.245	-2.000	0.000	1.000	2.000	3.000
wheelbase	205.000	98.757	6.022	86.600	94.500	97.000	102.400	120.900
carlength	205.000	174.049	12.337	141.100	166.300	173.200	183.100	208.100
carwidth	205.000	65.908	2.145	60.300	64.100	65.500	66.900	72.300
carheight	205.000	53.725	2.444	47.800	52.000	54.100	55.500	59.800
curbweight	205.000	2555.566	520.680	1488.000	2145.000	2414.000	2935.000	4066.000
enginesize	205.000	126.907	41.643	61.000	97.000	120.000	141.000	326.000
boreratio	205.000	3.330	0.271	2.540	3.150	3.310	3.580	3.940
stroke	205.000	3.255	0.314	2.070	3.110	3.290	3.410	4.170
compressionratio	205.000	10.143	3.972	7.000	8.600	9.000	9.400	23.000
horsepower	205.000	104.117	39.544	48.000	70.000	95.000	116.000	288.000
peakrpm	205.000	5125.122	476.986	4150.000	4800.000	5200.000	5500.000	6600.000
citympg	205.000	25.220	6.542	13.000	19.000	24.000	30.000	49.000
highwaympg	205.000	30.751	6.886	16.000	25.000	30.000	34.000	54.000
price	205.000	13276.711	7988.852	5118.000	7788.000	10295.000	16503.000	45400.000

The descriptive statistics show considerable variation among the automobile features. Price also has a wide range, indicating substantial differences in the market values of the vehicles.

4.2 Categorical Feature Analysis

```
fueltype:
['gas' 'diesel']
Number of unique values: 2

aspiration:
['std' 'turbo']
Number of unique values: 2

doornumber:
['two' 'four']
Number of unique values: 2

carbody:
['convertible' 'hatchback' 'sedan' 'wagon' 'hardtop']
Number of unique values: 5

drivewheel:
['rwd' 'fwd' '4wd']
Number of unique values: 3

enginelocation:
['front' 'rear']
Number of unique values: 2

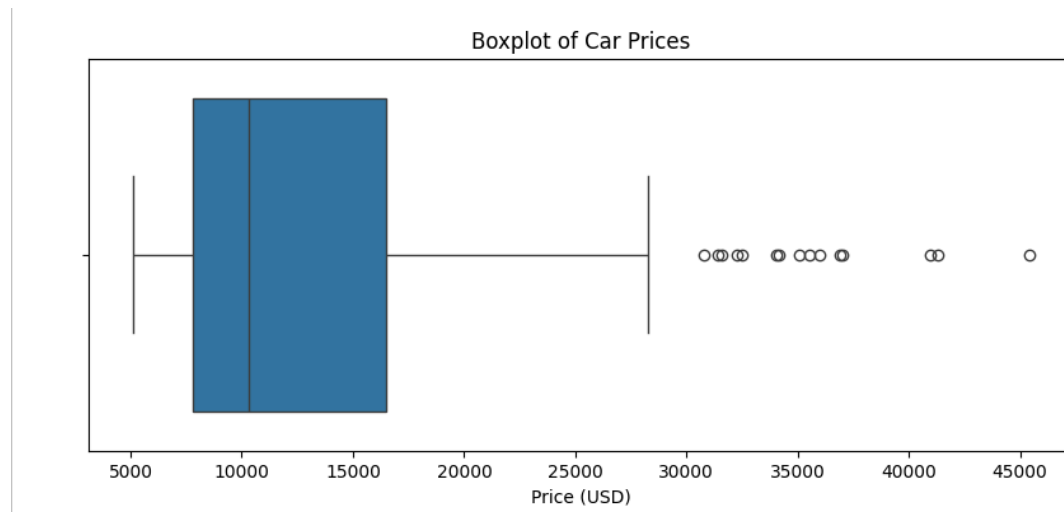
enginetype:
['dohc' 'ohcv' 'ohc' 'l' 'rotor' 'ohcf' 'dohcv']
Number of unique values: 7

cylindernumber:
['four' 'six' 'five' 'three' 'twelve' 'two' 'eight']
Number of unique values: 7

fuelsystem:
['mpfi' '2bbl' 'mfi' '1bbl' 'spfi' '4bbl' 'idi' 'spdi']
Number of unique values: 8
```

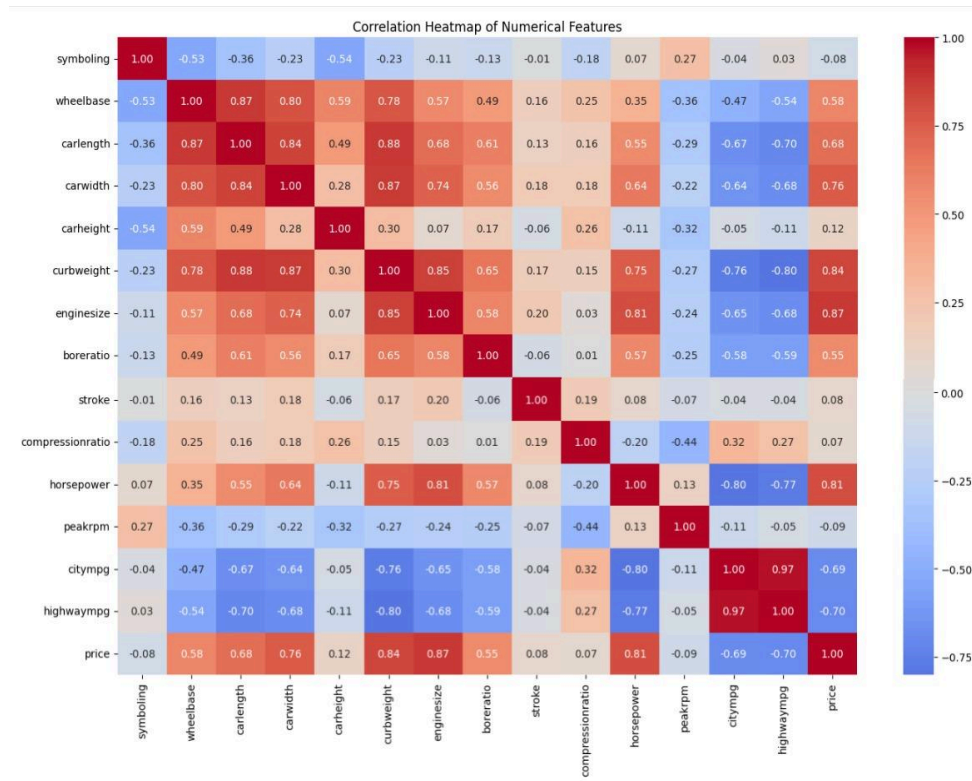
The categorical features represent different characteristics of the automobiles, including fuel type, body type, drivetrain, engine configuration, and fuel system. These variables were considered during preprocessing and model development.

4.3 Outlier Analysis



The boxplot shows that most car prices fall within a relatively lower range, while several higher-priced cars appear beyond the upper whisker. This agrees with the IQR analysis, which identified 15 potential outliers. These values were retained because they may represent genuinely expensive vehicles rather than incorrect observations.

4.4 Correlation Analysis



The correlation heatmap helps understand how different numerical features correlate with the car price. Based on the heatmap, it could be concluded that such variables as engine size, curb weight, and horsepower have a strong positive correlation with the price. Hence, the bigger those values are, the higher the car price tends to be. In contrast, the city and highway mileage has a negative correlation with the car price, meaning that higher mileage results in lower prices. Lastly, the heatmap also represents strong correlations between some of the car attributes which should also be considered while selecting features for the regression model.

V MODEL

Feature Selection

	Feature	VIF
3	carwidth	2828.145
1	wheelbase	2254.717
2	carlength	1893.738
4	carheight	929.458
13	highwaympg	511.133
12	citympg	437.334
5	curbweight	403.472
7	boreratio	291.357
11	peakrpm	219.619
8	stroke	126.412
6	enginesize	68.872
10	horsepower	66.166
9	compressionratio	15.835
0	symboling	2.541

Selected Numerical Features:
['curbweight', 'horsepower', 'carwidth', 'wheelbase']

Selected Categorical Features:
['carbody', 'enginelocation', 'manufacturer']

Fig.5.12 : Selected numerical features

Fig.5.11 : VIF table

After looking at the correlations between the numerical variables, VIF was used to get a clearer idea of how much the predictors overlap with each other. The results showed very high VIF values for several vehicle-related features, particularly carlength, wheelbase and carwidth. Even though there was considerable multicollinearity, the selected numerical features were retained for the modelling stage because they describe different aspects of the vehicle and can contribute useful information when predicting price.

The final numerical features selected for the model were curbweight, horsepower, carwidth, and wheelbase. The categorical features selected were carbody, enginelocation, and manufacturer. These features were used in the final Multiple Linear Regression model.

VI Model Evaluation

6.1 Regression Assumption Checking

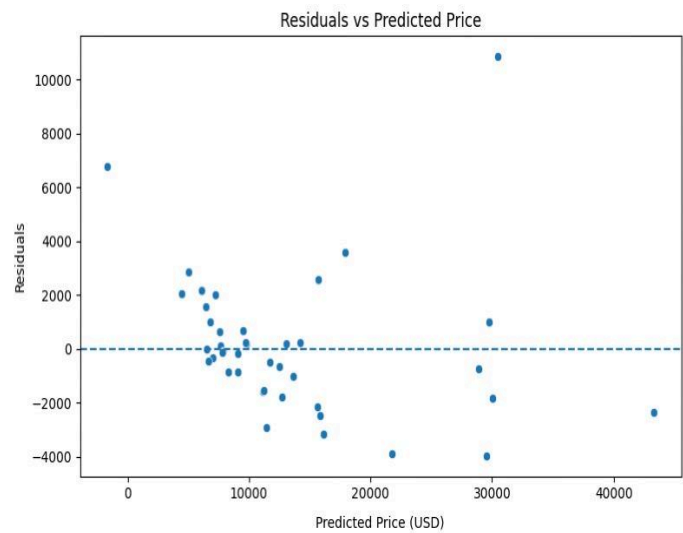


Fig 6.11 : Residuals

	0
LM Statistic	95.695
LM-Test p-value	0.005
F-Statistic	2.470
F-Test p-value	0.000
dtype:	float64

Fig 6.12 : Breusch-Pagan

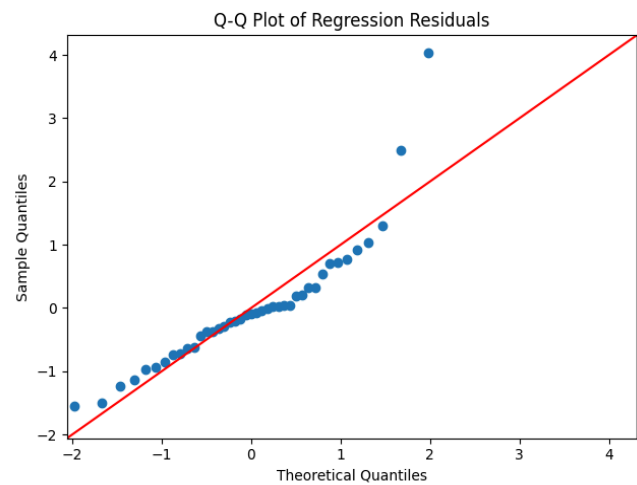


Fig 6.13 : Homoscedasticity

The residuals were plotted against the predicted prices to see whether there was any clear pattern in the model errors. Most of the points are distributed around the zero line, although the spread is not completely even, especially for some of the higher predicted prices. This suggests that the model captures the general relationship reasonably well, but there may still be some variation in the error size across different price ranges. The Breusch–Pagan test also indicated the presence of heteroscedasticity, suggesting that the variance of the errors is not completely constant.

6.2 Performance Evaluation

```
Selected Numerical Features:  
['curbweight', 'horsepower', 'carwidth', 'wheelbase']
```

```
Selected Categorical Features:  
['carbody', 'enginelocation', 'manufacturer']
```

```
--- Test Set Performance ---
```

```
MAE : 2229.24  
MSE : 10229416.98  
RMSE : 3198.35  
R2 : 0.8704
```

```
--- Training Set Performance ---
```

```
MAE : 1108.74  
MSE : 2349239.95  
RMSE : 1532.72  
R2 : 0.9606
```

```
Number of training samples: 164
```

```
Number of testing samples: 41
```

Final performance output:

Metric	Training	Testing
MAE	1108.74	2229.24
MSE	2,349,239.95	10,229,416.98
RMSE	1532.72	3,198.35
R ²	0.9606	0.8704

Model Performance Observation

The final model was tested on both the training and testing data to see how well the selected features could predict car prices. It achieved an R^2 value of 0.9606 on the training data and 0.8704 on the testing data, which shows that the model explains a large portion of the variation in car prices. The RMSE on the testing data was 3198.35, meaning that the prediction errors were higher for unseen cars than for the training data. Overall, the model gives reasonably good predictions, although the difference between the training and testing results shows that its performance drops slightly when working with new data.

6.3 Statistical Significance of Predictors

PREDICTOR	P-VALUE	SIGNIFICANCE
Curbweight	<0.001	Yes
Horsepower	<0.001	Yes
Carwidth	0.002	Yes
Wheelbase	0.361	No
Carbody_hatchback	0.002	Yes
Carbody_sedan	0.002	Yes
Carbody_wagon	<0.001	Yes
Enginelocation_rear	<0.001	Yes

The OLS results were used to check which of the selected variables had a statistically significant relationship with car price. Using a 5% significance level, curbweight, horsepower, and carwidth were found to be significant numerical predictors, while wheelbase was not statistically significant in the final model. Among the categorical variables, several car body types and the rear engine location showed significant effects on price. Some manufacturer categories, including BMW, Buick, Jaguar, Mitsubishi, Peugeot, Porsche, and Toyota, were also statistically significant. Overall, the results suggest that vehicle weight, engine power, dimensions, body type, engine location, and certain manufacturers contribute meaningfully to differences in car prices.

VII Results & Discussion

Model Interpretation

The final regression model shows that several vehicle characteristics have a noticeable influence on car prices. Features such as curbweight, horsepower, and carwidth were statistically significant, indicating that differences in these characteristics are associated with changes in the predicted price. The categorical variables also provided useful information, particularly car body type, engine location, and some manufacturers. Although wheelbase was included in the final model, it was not statistically significant at the 5% level. Overall, the results show that car prices are influenced by a combination of physical characteristics and categorical factors rather than by a single variable.

Limitations

Although the final model gives reasonably good predictions, there are some limitations to consider. The difference between the training and testing performance suggests that the model performs better on the data it was trained on than on unseen data. The diagnostic analysis also indicated that the error variance is not completely constant. In addition, some predictors showed high multicollinearity during the analysis, which can make it harder to interpret the individual effect of closely related variables. These limitations do not make the model unusable, but they should be considered when applying the results to other car price data.

Conclusion

This analysis developed a multiple linear regression model to predict car prices using a combination of numerical and categorical features. After exploring the dataset, checking for outliers and multicollinearity, and selecting relevant features, the final model achieved an R^2 of 0.8704 on the testing data. The results show that factors such as curbweight, horsepower, carwidth, car body type, engine location, and certain manufacturers can contribute to differences in car prices. While the model has some limitations, particularly in terms of error variance and the difference between training and testing performance, it provides a useful approach for understanding and predicting car prices from the available vehicle characteristics.

TASK 3

Bank Customer Churn Prediction -Logistic Regression

I Domain Explanation

The banking industry handles a large amount of customer information, including personal details, account activity, and financial behaviour. Understanding this information can help banks identify customers who may stop using their services. Customer churn can be influenced by factors such as age, credit score, account balance, tenure, number of products, and customer activity.

In this study, Logistic Regression is used to predict whether a bank customer is likely to churn based on the available customer and account-related features. Exploratory Data Analysis (EDA) is first carried out to understand the data and identify patterns related to customer churn before developing and evaluating the classification model.

II Problem Statement

The aim of this study is to develop a Logistic Regression model to predict whether a bank customer will churn based on the available customer and account-related features.

The study focuses on understanding the factors associated with customer churn, preparing the data for classification, building a Logistic Regression model, and evaluating its performance using suitable classification metrics. The model will help determine how effectively customer characteristics can be used to identify customers who are likely to leave the bank.

Target Variable: Churn

III Dataset Explanation

The dataset contains information about 10,000 bank customers with 12 variables describing their personal details, credit information, account characteristics, and banking activity. The dataset contains both numerical and categorical variables. The categorical variables are country and gender, while the remaining variables are numerical.

The churn variable is used as the target variable, where 0 represents customers who did not churn and 1 represents customers who churned. The customer_id variable is used only as an identifier and is not included as a predictor in the model.

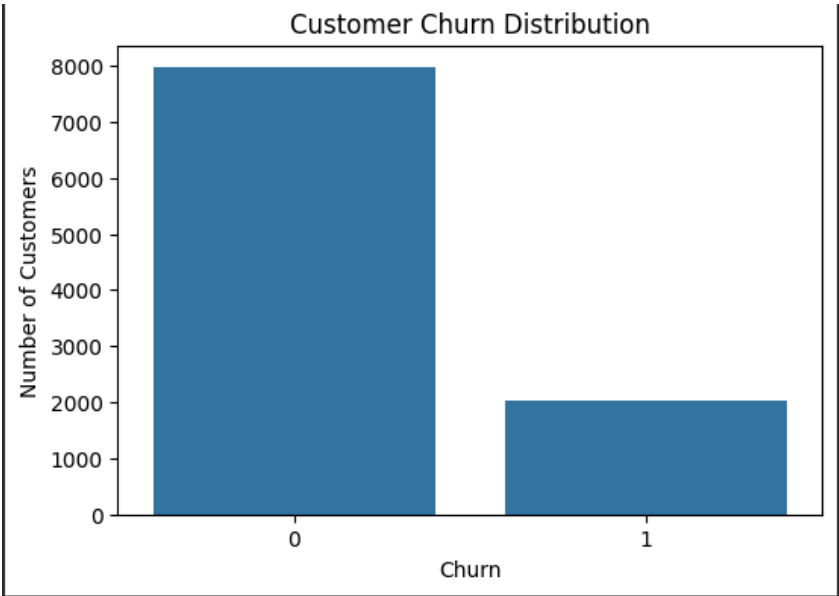
The dataset contains no missing values and no duplicate records. Among the 10,000 customers, 7,963 did not churn and 2,037 customers churned.

Dataset Summary

CATEGORY	DETAILS
Number of records	10,000
Number of variables	12
Target variable	churn
Numerical variables	Credit score, age, tenure, balance, products number, credit card, active member, estimated salary
Categorical variables	Country, gender
Missing values	0
Duplicate records	0
Churned customers	2,037
Non-churned customers	7,963

IV. Exploratory Data Analysis (EDA)

4.1 Churn Distribution

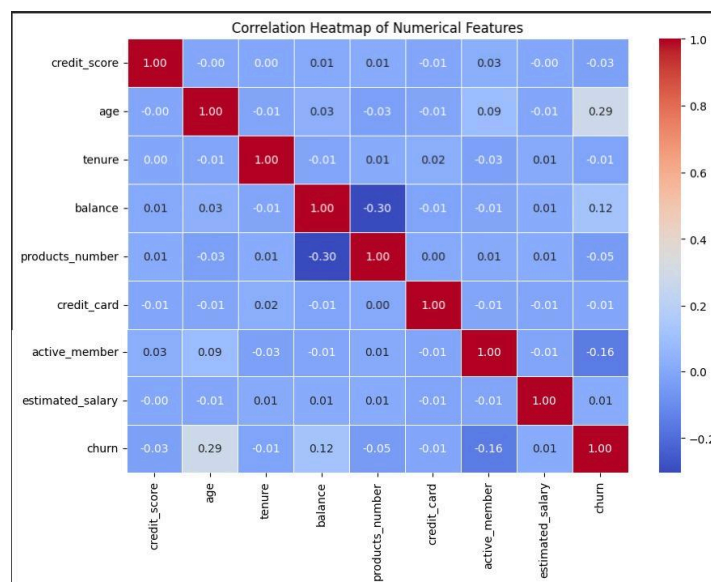


OBSERVATION:

The churn distribution shows that most customers did not leave the bank. Out of 10,000 customers, 7,963 customers (79.63%) did not churn, while 2,037 customers (20.37%) churned. This indicates that the dataset is imbalanced towards non-churned customers. Therefore, evaluation metrics such as precision, recall, F1-score, and ROC-AUC are important in addition to accuracy .

4.2 Numerical Feature Analysis

CORRELATION HEATMAP



The correlation heatmap was used to examine the relationships between the numerical features and the churn variable. It provides an overview of the strength and direction of relationships among the numerical variables and helps identify features that may be useful for predicting customer churn.

V. Model

The dataset was prepared by separating the predictor variables from the target variable, churn. The customer_id column was excluded because it is only an identifier and does not provide useful predictive information. The categorical variables, country and gender, were converted into numerical dummy variables using one-hot encoding, with the first category dropped to avoid redundancy. The dataset was divided into training and testing sets, and the numerical features were standardized using StandardScaler.

Logistic Regression was then used as the classification model to predict whether a customer would churn or remain with the bank. The model estimates the probability of customer churn based on the available customer and account-related features and classifies customers into churn and non-churn categories.

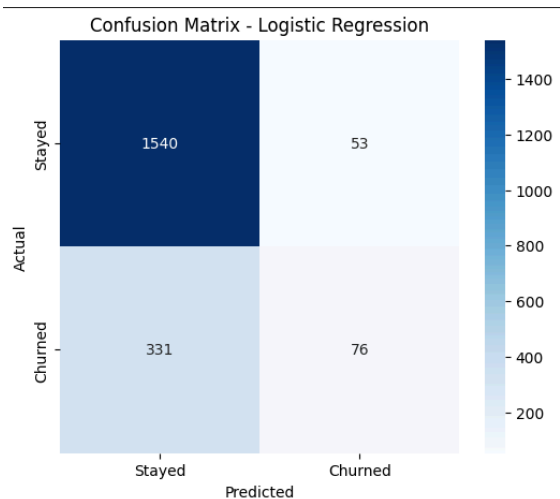
VI. Model Evaluation

6.1 Performance Evaluation

METRIC	RESULT
Accuracy	80.80%
Presicion	0.5891
Recall	0.1867
F1-score	0.2836
ROC-AUC	07748

The Logistic Regression model achieved an accuracy of 80.80% on the testing data. The precision was 0.5891, while the recall was 0.1867 and the F1-score was 0.2836. The ROC-AUC score was 0.7748, indicating that the model has a reasonable ability to distinguish between customers who churn and those who do not. The lower recall shows that the model was not able to identify a large proportion of the actual churned customers.

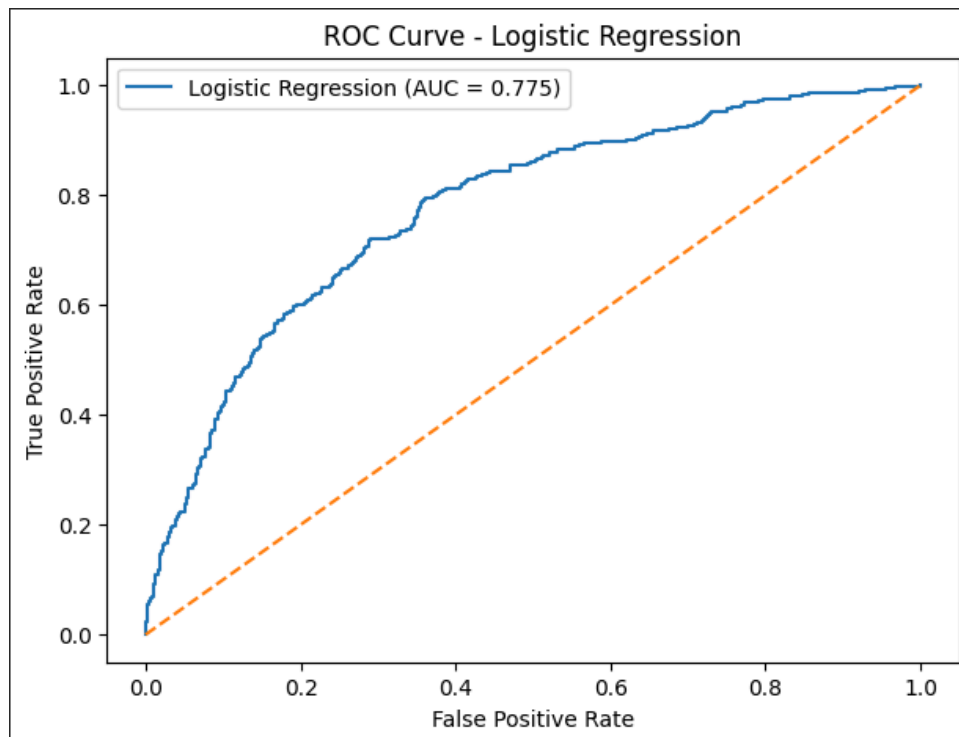
6.2 Confusion Matrix



OBSERVATION:

The confusion matrix shows that the model correctly classified 1,540 non-churned customers and 76 churned customers. It incorrectly classified 53 non-churned customers as churned and 331 churned customers as non-churned. The relatively high number of false negatives explains the lower recall of the model.

6.3 ROC Curve



The ROC curve was used to evaluate how well the model distinguishes between churned and non-churned customers at different classification thresholds. The model achieved a ROC-AUC score of 0.7748, showing a reasonable level of discriminatory ability.

VII. Results & Discussion

Model Interpretation

The Logistic Regression model was able to predict customer churn with an accuracy of 80.80% on the testing data. The ROC-AUC score of 0.7748 shows that the model has a reasonable ability to distinguish between customers who are likely to churn and those who are not. However, the recall value of 0.1867 is relatively low, which means that the model was unable to identify many of the customers who actually churned. The confusion matrix also shows a higher number of false negatives compared with true positives. Therefore, although the model provides a useful baseline for churn prediction, there is scope for improving its ability to identify potential churn customers.

Limitations

The main limitation of the model is its low recall, which means that a considerable number of actual churn customers were classified as non-churn customers. The dataset also contains more non-churned customers than churned customers, which can affect the classification results. In addition, Logistic Regression assumes a relatively simple relationship between the predictors and the probability of churn. More advanced classification techniques or further feature engineering could potentially improve the model's ability to identify customers who are likely to churn.

Conclusion

This study developed a Logistic Regression model to predict customer churn using customer and account-related features. The dataset was explored and prepared before training the model, and its performance was evaluated using accuracy, precision, recall, F1-score, confusion matrix, and ROC-AUC. The model achieved an accuracy of **80.80%** and a ROC-AUC score of **0.7748** on the testing data. However, the recall of **0.1867** indicates that identifying actual churn customers remains a challenge. Overall, the model provides a useful starting point for understanding customer churn and can be further improved using suitable feature engineering and classification techniques.